

Topology

Home Work 2

Student: 26-712-224

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Exercise 1

- i) Let $X = \mathbb{R}$ and prove that $\mathcal{U} = \{(-\infty, a) \mid a \in \mathbb{R} \cup \{\pm\infty\}\}$ defines a topology on X .
- ii) Let $X = \mathbb{R}$ and prove that $\mathcal{U} = \{(-\infty, a] \mid a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}$ does not define a topology on X .
- iii) Let X be a set, define $\mathcal{T} = \{A \mid X \setminus A \text{ is finite}\} \cup \{\emptyset\}$; prove that this is a topology (this is called the cofinite topology).
- iv) List 4 topologies on the set of three elements $\{a, b, c, d\}$ that have different numbers of open sets.

Answer:

In each of the 4 cases above we must either show that the 3 conditions for a topology are met OR find at least one condition which is not met.

(i) Condition $\emptyset, X \in \mathcal{U}$: Both $\emptyset = (-\infty, -\infty) \in \mathcal{U}$ and also $\mathbb{R} = X = (-\infty, \infty) \in \mathcal{U}$. This condition is satisfied.

Union condition: Let $(U_i = (-\infty, a_i))_{i \in I}$ be a collection of open sets in $\mathcal{U}$. Now consider the union:

$$\begin{aligned} J &= \bigcup_{i \in I} U_i \\ &= \bigcup_{i \in I} (-\infty, a_i) \end{aligned} \tag{1.1}$$

In (1.1) if any $U_i = (-\infty, \infty)$ then, clearly: $J = (-\infty, \infty) \in \mathcal{U}$. If all $a_i < \infty$, then consider the set $A = \{a_i \mid i \in I\} \subset \mathbb{R}$. Let $a = \text{Sup}(A) \in (\mathbb{R} \cup \infty)$ (as: $a = \infty$ possible). Then it should be clear that: $J = (-\infty, a) \in \mathcal{U}$.

Hence the union of any collection of elements from $\mathcal{U}$ is again in $\mathcal{U}$.

Intersection condition: Let $(U_i = (-\infty, a_i))_{i=1}^m$ be a finite collection of open sets from $\mathcal{U}$. In this case let: $a = \min_i(a_i)$.

Then it is clear that:

$$\begin{aligned}
I &= \bigcap_{i=1}^m U_i \\
&= \bigcap_{i=1}^m (-\infty, a_i) \\
&= (-\infty, a)
\end{aligned}$$

Hence, the intersection of a finite number of open sets from $\mathcal{U}$ is again open. So we see that all 3 conditions for a topology are met.

(ii) The "union condition" is violated. Let $(U_i = (-\infty, 2 - \frac{1}{i}])_{i \in \mathbb{N}_+}$. Then:

$$\begin{aligned}
J &= \bigcup_{i \in \mathbb{N}_+} U_i \\
&= \bigcup_{i \in \mathbb{N}_+} (-\infty, 2 - \frac{1}{i}] \\
&= (-\infty, 2)
\end{aligned}$$

But $J = (-\infty, 2) \notin \mathcal{U}$ and hence $\mathcal{U}$ is not a valid topology on $\mathbb{R}$.

(iii) Condition $\emptyset, X \in \mathcal{T}$: By definition $\emptyset \in \mathcal{T}$. Also $X^c = \emptyset$ which is a finite set and hence $X \in \mathcal{T}$.

Union condition: Let $(U_i)_{i \in I}$ be a collection of open sets in $\mathcal{T}$. And let:

$$J = \bigcup_{i \in I} U_i$$

If every $U_i = \emptyset$ then $J = \emptyset \in \mathcal{T}$. Otherwise $J \neq \emptyset$ and we can take the complement of J :

$$\begin{aligned}
J^c &= \left[\bigcup_{i \in I} U_i \right]^c \\
&= \bigcap_{i \in I} U_i^c
\end{aligned} \tag{1.2}$$

As, in this case, at least one of the $U_i \neq \emptyset$, let one be U_k (for some $k \in I$). From (1.2): $J^c \subset U_k^c$ and as U_k^c is finite we must conclude that J^c is finite and hence $J \in \mathcal{T}$ - therefore the union condition is met.

Intersection condition: Let $(U_i)_{i=1}^m$ be a finite collection of open sets from $\mathcal{T}$. Let:

$$I = \bigcap_{i=1}^m U_i \tag{1.2}$$

If **any** $U_i = \emptyset$, then $I = \emptyset \in \mathbb{U}$ - nothing more to prove. Otherwise, $I \neq \emptyset$ and taking the complement of I :

$$\begin{aligned} I^c &= \left[\bigcap_{i=1}^m U_i \right]^c \\ &= \bigcup_{i=1}^m U_i^c \end{aligned}$$

As **all** the U_i in this union are not empty - the union is effectively the finite union of sets each of which are finite. It must therefore be the case that I^c is also finite - i.e. that $I \in \mathcal{T}$ - therefore the intersection condition is met.

(iv) Given $X = \{a, b, c, d\}$ we can define 4 structurally different topologies:

- (a) $\mathcal{T}_1 = \{\emptyset, X\}$
- (b) $\mathcal{T}_2 = \{\emptyset, X, \{a\}\}$
- (c) $\mathcal{T}_3 = \{\emptyset, X, \{a\}, \{a, b\}\}$
- (d) $\mathcal{T}_4 = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

Exercise 3

Let $X = \mathbb{R}$ be the real numbers and $\mathcal{B} = \{[a, b) \mid a, b \in \mathbb{R}\}$.

- i) Check that $\mathcal{B}$ is a base of topology.
- ii) Let $\mathcal{U} = \mathcal{U}(\mathcal{B})$ be the topology generated by $\mathcal{B}$. Show that if a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on the right then $f : (\mathbb{R}, \mathcal{U}) \rightarrow (\mathbb{R}, \mathcal{U})$ is continuous.

Reminder: f is continuous on the right if for all $x \in X$ and all $\epsilon > 0$ there exists $\delta > 0$ such that if $x \leq x' < x + \delta$ then $|f(x) - f(x')| < \epsilon$.

Answer:

(i) What I think this question is asking is: "Can $\mathcal{B}$ generate a topology on X ?". For this to be the case we have two conditions to be met:

$$\forall x \in \mathbb{R}, \exists B \in \mathcal{B} \text{ such that } x \in B \quad (3.1)$$

And:

$$B_1, B_2 \in \mathcal{B} \text{ with } x \in B_1 \cap B_2 \implies \exists B_3 \in \mathcal{B} \text{ s.t. } x \in B_3 \subset B_1 \cap B_2 \quad (3.2)$$

For any $x \in \mathbb{R}$ we can take: $B = [-[x] - 1, [x] + 1) \in \mathcal{B}$ - where $|x|$ is the absolute value of x and $[x]$ is the integer value of x . Hence, $x \in B$ and condition (3.1) is met.

If $B_1 = [a, b)$, $B_2 = [c, d) \in \mathcal{B}$ and $x \in B_1 \cap B_2$ then we have that: $B_1 \cap B_2 = [m, n)$, where $m = \max(a, c)$ and $n = \min(b, d)$. So, we can take $B_3 = [m, n) \in \mathcal{B}$ and with this condition (3.2) is satisfied.

Hence, $\mathcal{B}$ satisfies the conditions needed to be the generator for a topology.

(ii) The diagram below shows a simple function that is right-continuous:

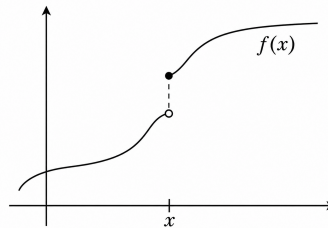


Figure 1: Right-continuous function with a jump discontinuity at x .

The black "blob" in the graph is the point $(x, f(x))$.

To prove continuity of such a function using the $(\mathbb{R}, \mathcal{U})$ topology we just need to show that if we take any $B \in \mathcal{U}$, then $f^{-1}(B) \in \mathcal{U}$.

I am at a loss as to the next step!

Exercise 5

Let A be a subset of a topological space X . Show that:

- i) $A^\circ = A$ if and only if A is open in X .
- ii) $\bar{A} = A$ if and only if A is closed in X .
- iii) $X \setminus \bar{A} = (X \setminus A)^\circ$.

Answer:

(i) $A^\circ = A \iff A$ is open.

(a) $A^\circ = A \implies A$ is open. By definition:

$$A^\circ = \bigcup_{O \subset A} O \subset A \quad \text{where } O \text{ is open} \quad (5.1)$$

From (5.1) we see that A° is defined as the union of a collection of open sets - and so must be open. As $A^\circ = A$, then A must be open.

(b) A open $\implies A^\circ = A$. Well this is straight forward! A will be one of the O in the union of (5.1).

(ii) $\bar{A} = A \iff A$ is closed.

(a) $\bar{A} = A \implies A$ is closed. By definition:

$$A \subset \bar{A} = \bigcap_{C \supset A} C \quad \text{where } C \text{ is closed and contains } A \quad (5.2)$$

From (5.2) $\bar{A}$ is the intersection of a collection of closed sets and so must itself be closed. With $\bar{A} = A$, we must conclude that A is closed.

(b) A closed $\implies \bar{A} = A$. If A is closed it will be one of the $C \supset A$ in the intersection and then $\bar{A} = A$.

(iii) $[\bar{A}]^c = (A^c)^\circ$.

From the definitions we know that: $A^\circ \subset A \subset \bar{A}$.

If we now take the complements: $(A^\circ)^c \supset A^c \supset [\bar{A}]^c \implies (A^c)^\circ \supset [\bar{A}]^c$ (*), since $[\bar{A}]^c$ must be open and $(A^c)^\circ$ is the largest open set in (A^c) .

Now assume that $x \in (A^c)^\circ$, that means there is an open set U such that: $x \in U \subset (A^c)$, i.e. $U \cap A = \emptyset$ and hence $x \notin \bar{A}$. That is to say $x \in [\bar{A}]^c$ and hence:

$$(A^c)^\circ \subset [\bar{A}]^c \text{ (**).}$$

The two way inclusions (*) and (**) prove the required equality.

Exercise 6

Let X be an uncountable set; define the cocountable topology on X as

$$\mathcal{U}_{\text{cocountable}} = \{U \subseteq X; X \setminus U \text{ is finite or countable}\} \cup \{\emptyset\}.$$

- i) Prove that one may not find a countable base for topology $\mathcal{U}_{\text{cocountable}}$ on X .
- ii) Show that if $x_n \rightarrow x$ (in the cocountable topology), then the sequence x_n stabilizes to x (i.e., there exists $N \in \mathbb{N}$ such that $x_n = x$ for all $n > N$).

Recall the definition of sequentially convergence in topology: $x_n \rightarrow x$ in topology $(X, \mathcal{U})$, if for every neighborhood U of x in $(X, \mathcal{U})$, there exists $N \in \mathbb{N}$ such that $x_n \in U$ for all $n > N$.

- iii) Consider the identity function $I_{\mathbb{R}} : (\mathbb{R}, \mathcal{U}_{\text{cocountable}}) \rightarrow (\mathbb{R}, \mathcal{U}_{\mathbb{R}})$, where $\mathcal{U}_{\mathbb{R}}$ denotes the Euclidean topology induced by the metric $d(x, y) = |x - y|$. Prove that $I_{\mathbb{R}}$ is not continuous; however, the image of every converging sequence converges (i.e., $I_{\mathbb{R}}$ is sequentially continuous.)

Answer:

Note: I use $\mathcal{U}$ for $\mathcal{U}_{\text{cocountable}}$.

(i) Let

$$B \subset \mathcal{U} = \{u \subset X \mid X - u \text{ is finite/countable}\} \cup \{\emptyset\}$$

be a base for $\mathcal{U}$.

Then it is clear that each $b \in B$ must be uncountable. Now assume that B itself is countable - i.e.

$$B = (b_i)_{i \in \mathbb{N}} \quad (\text{and } \{\emptyset\} \notin B \text{ for simplicity})$$

From each b_i select an element c_i .

Then $(c_i)_{i \in \mathbb{N}}$ is a sequence that contains at least one element from each b_i .

Define

$$C = \{c_1, c_2, \dots\}$$

which is countable.

As:

$$C = X \setminus (X \setminus C)$$

It is clear that $X \setminus C$ is open and hence $(X - C) \in \mathcal{U}$.
 But, from the definition of a base for $\mathcal{U}$, there is an index $I \subset \mathbb{N}$ s.t

$$X \setminus C = \bigcup_{i \in I} b_i$$

This is a contradiction as $X \setminus C$ does not contain the $(c_i)_{i \in I}$.

□

(ii) Let $x_n \rightarrow x$. Then for any $U \in \mathcal{U}$, $\exists N \in \mathbb{N}$ s.t.:

$$x_n \in U, \forall n > N \quad \& \quad x \in U$$

Assume that (x_n) never stabilizes to x .

Let:

$$D = \{x_n \mid x_n \neq x\}$$

and D is countably infinite.

Hence: $(X \setminus D) \in \mathcal{U}$, i.e. we have found an open set in $\mathcal{U}$ s.t.

$$x \in (X \setminus D) \in \mathcal{U}$$

but, by the definition of D :

$$\nexists N \in \mathbb{N} \text{ s.t.}$$

$$x_n \in (X \setminus D) \quad \forall n > N$$

Contradiction. Hence x_n must stabilize to x at some finite $n \in \mathbb{N}$.

□

(iii) The identity function: $f : (\mathbb{R}, \mathcal{U}) \rightarrow (\mathbb{R}, \mathcal{U}_{\mathbb{R}})$, where $f(x) = x$, with the two given topologies is not continuous.

To demonstrate this we can take an open set in $\mathcal{U}_{\mathbb{R}}$ and show that its inverse image is NOT an open set in $\mathcal{U}$.

Take $(0, 1) \in \mathcal{U}_{\mathbb{R}}$, clearly $f^{-1}((0, 1)) = (0, 1)$. However, $\mathbb{R} \setminus (0, 1)$ is not a countable set - and hence not open in $\mathcal{U}$.

□